Faithful symmetric lifts on pairs

Human Compression project: a component of the L044 recovery

Let $G = A_{2m} = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be naturally permuted and have a string C-marking. Suppose its maximal omission $H = G_i$ is transitive on $m \geq 8$ pairs, with block image S_m. Write $V = \mathbb{F}_2^m$, let z be the constant flip, and let E be the even-weight subspace. Choose the minimal distinguished block-generating family L, with $B = \langle L \rangle$, and the commuting and remaining families C, R. We acquire the full L047 decomposition and the accepted actual pair-centralizer lemma. In any putative case $r \geq m$, L is an interval, $|C| \leq 1$ and $|R| \leq 1$.

Lemma 1. *Suppose a permutation group preserves one pair setwise and induces S_t on its other $t \geq 4$ pairs. Every permutation centralizing that group preserves all its pairs.*

Proof. Its point orbits on the other pairs have size at least t, whereas those in the distinguished pair have size at most two. A centralizer preserves these two unions of orbits. On the $2t$ points the acquired pair-centralizer lemma applies: the block group S_t is primitive and is not a regular prime-dihedral group of order $2t$ for $t \geq 4$. It forces preservation of every pair there. The remaining two points are the distinguished pair. No faithfulness of the original group's restriction to either union is presumed; the lemma is applied to its actual induced permutation group. □

Lemma 2. *A connected string C-marking of the natural S_m, of rank $m - 2$, has, up to simultaneous conjugacy and reversal, the form*

$$(2\ 3), \quad (1\ 2)(3\ 4), \quad (4\ 5), \quad (5\ 6), \dots, (m-1\ m), \qquad m \geq 8. \tag{1}$$

Its tail-end omission is the natural S_{m-1} fixing m; its core-end omission is $S_{\{1,2\}} \times S_{\{3,\dots,m\}}$.

Proof. For $m \geq 9$ we acquire the complete B5 classification, including its general four-point cores and attached trees. Its core B produces a triangle of noncommutation with the first attached edge, and therefore has no string order. In core C, that first attached edge fails to commute with both core marks and with the next attached edge; its degree in the noncommutation graph is at least three. There is such a next edge because the root is a leaf and the tail has at least two edges. This also cannot be a string. In core A, the first two marks are the first two entries of (1). The attached tree has no vertex of degree three, since its incident transpositions would give a triangle of noncommutation. Hence it is a path. The resulting noncommutation diagram is itself a path, so its only string orders are its path order and reversal, giving (1).

The degree-eight boundary is a complete new execution of the acquired L047 prefix-stabilizer engine, with the actual ambient group changed to S_8 and target and depth both six. The search includes every nonidentity involution, requires independence and the string commutations, and requires noncommutation at each consecutive position. It takes orbits only under the stabilizer of the entire prefix. Every discarded prefix either already has a redundant mark, fails a necessary end-intersection equality, or has insufficient subgroup index or centralizer quotient index for the remaining independent marks. These are exactly the retained completeness conditions of that engine. In particular, the centralizer-quotient test is applied only to a possible completion with full IP: the subgroup on the future marks, except the immediate next mark, intersects the old prefix group trivially and so injects into its centralizer modulo the old center.

All rank-six candidates are tested against the full intersection property. The complete run has prefix counts

$$1, 4, 63, 115, 22, 9, 3$$

and exactly two surviving ordered conjugacy classes, with types $[4, 6, 3, 3, 3]$ and $[3, 3, 3, 6, 4]$. A separate exact check verifies generation of literal S_8, all involutions and string commutations, all 64^2 subset intersections for each tuple, and simultaneous conjugacy to (1) or its reverse. It also checks that these two ordered classes differ. Thus the same conclusion holds at eight. This is only the stated rank-six connected boundary, not a census of every rank or every C-group.

The tail omission in (1) fixes m and generates S_{m-1}: the core with its first attachment generates S_5, and each further transposition attaches one new point. For the other omission, if $u = (1\ 2)(3\ 4)$ and $v = (4\ 5)$, then $(uv)^3 = (1\ 2)$. This recovers $(3\ 4)$ and all of the remaining path transpositions. The generated group is exactly the displayed product. □

Lemma 3. *A subgroup of $C_2 \wr S_m$ projecting isomorphically to S_m, for $m \geq 8$, is, after coordinates are chosen in the pairs, either the ordinary doubled action or*

$$\pi : (j, \epsilon) \longmapsto (j^\pi, \epsilon + \operatorname{sgn}_2 \pi). \tag{2}$$

The second action is excluded for a subgroup of A_{2m} when m is odd.

Proof. If the subgroup has two point orbits, each meets each pair once; these two rows give the ordinary doubled action. Otherwise it is transitive, and a point stabilizer has index two in a block stabilizer S_{m-1}. Its unique index-two subgroup is A_{m-1}. The coset action on S_m/A_{m-1} is exactly (2); an equivariant identification can be chosen to match the specified pairs. In (2) an odd π is its ordinary doubled permutation, which is even, multiplied by the flip of all m pairs. For odd m this is an odd permutation, as claimed. □

Theorem 4. *Under the standing hypotheses, a putative rank $r \geq m$ can have neither $|L| = m-1$ nor $B \cap V = 1$. Thus it forces*

$$r = m, \quad |L| = m - 2, \quad C = \varnothing, \quad |R| = 1, \quad B \cap V = \langle z \rangle, \quad H \cap V = E, \quad m \text{ even}. \tag{3}$$

Proof. The invariant subspaces of the permutation module are $0, \langle z \rangle, E, V$, and $H \cap V \leq E$. If C is nonempty, its mark is z and m is even. IP excludes z from B, so the same subspace list gives $B \cap V = 1$. If R were also nonempty, $\langle L, R \rangle$ would strictly contain B and thus have nonzero kernel. For even m that kernel contains z, contrary to IP with the mark in C. Consequently

$$|C| + |R| \leq 1, \qquad r \leq |L| + 2, \qquad m - 2 \leq |L| \leq m - 1. \tag{4}$$

If $|L| = m - 1$, its independent projected involutory string is the Coxeter path, by the complete acquired extremal theorem. The sign-modified alternative in that theorem cannot consist of involutions: an edge adjacent to its distinguished tree edge has order three. The omitted ρ_i commutes with the lifted subgroup obtained by removing the adjacent endpoint of L. That subgroup fixes one original pair setwise and induces S_{m-1} on the rest. Lemma 1 makes ρ_i preserve all pairs, contradicting the parent A_{2m}.

It remains that $|L| = m - 2$, $r = m$, and exactly one mark lies outside L in H. Suppose first that $B \cap V = 1$. Its faithful projection is a connected C-marking of S_m: L is an interval of the connected parent diagram, and faithfulness preserves its consecutive noncommutations. Lemma 2 applies. If the omitted ρ_i is next to the tail end, it centralizes the lifted tail omission and is excluded by Lemma 1. We may therefore orient (1) with ρ_i next to its core end. It centralizes the lifted core omission $S_2 \times S_b$, where $b = m - 2$. The pure A_b on the larger block orbit fixes exactly the four points over the small orbit in either action of Lemma 3. Thus ρ_i preserves those four points and the other $2b$ points separately.

If the one extra mark is $C = \{z\}$, then m and b are even. In the ordinary doubled action the centralizer on the $2b$ points is $\langle z_b \rangle$, whose elements are even. On the four points the core omission acts as a block-interchanging double transposition. Its even centralizer is the Klein four group, preserving the original pairs. Since ρ_i is even, its two restrictions have this parity and it preserves every pair. In the action (2), an odd element of the large S_b induces the full flip on the four small points, while the small S_2 induces the other block-interchanging double transposition. Their images generate the regular Klein four group there, whose centralizer is itself and preserves the original pairs. On the $2b$ points the pair-centralizer lemma, with block image S_b, again forces preservation of the pairs. This case is also impossible.

If instead the extra mark is in R, it is adjacent to the opposite tail end of L, because the omitted label occupies the core neighbour. It centralizes the lifted tail omission. Its block image centralizes the natural S_{m-1} fixing the tail point, hence is trivial. It is a flip vector constant off that endpoint. If m is even, evenness forces this vector to be z, which commutes with L and would belong to C, not R. Thus m is odd, the action of B is ordinary doubled, and the extra mark is $z + e_m$, flipping every pair except the tail pair. On the four core points it is their full flip, whereas the core omission supplies the block-interchanging double transposition. These generate a regular Klein four group. Since the omitted mark commutes with both, its restriction preserves the two original pairs there. On the remaining points it centralizes the doubled S_b, whose centralizer is $\langle z_b \rangle$ and preserves all its pairs. This again makes the parent imprimitive. We have excluded every faithful B.

Finally the module list leaves $B \cap V = \langle z \rangle$ or E. The latter would give $B = H$, contrary to the extra independent mark. Thus the kernel is $\langle z \rangle$ and m is even. The strict inclusion $B < H$ forces $H \cap V = E$. IP excludes $C = \{z\}$, so the extra mark belongs to R. Together with (4) this gives every assertion in (3). □

Attribution and finite premises. The rank route is source Section 3.2. The L047 decomposition, complete symmetric extremal theorem and B5 core classification remain acquired in their full reviewed scopes. The new work provides actual-cover centralizers, the parity argument for both faithful lifts, and the complete degree-eight string boundary. Its GAP engine changes only the ambient group and one target specification in the acquired L047 code. The guarded checker rejects optimized Python, clears that variable for children, pins inputs and checks actual tuples separately. GAP exact operations remain software premises. The nontrivial kernel case (3) is an explicit remaining obligation; no whole-original, open-question, Lean or compression credit is claimed by this component.